# Stat 578: Spectral Methods, Network Analysis, and Nonconvex Optimization
# HW 2

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**Due: 10/16 at 11:59PM**

1. Consider the model $\boldsymbol{P} = \boldsymbol{Z}\boldsymbol{B}\boldsymbol{Z}^\top$, where $\boldsymbol{Z}$ and $\boldsymbol{B}$ are as in the last homework. Suppose that one observes the "Gaussian" version of the model, i.e., we observe $\boldsymbol{A} = \boldsymbol{P} + \boldsymbol{N}$, where $\boldsymbol{N}$ is $\mathsf{GOE}(n)$ with variance $\sigma^2$. (This means that $\boldsymbol{N}_{ij} \sim \mathcal{N}(0, \sigma^2/2)$, $\boldsymbol{N}_{ii} \sim \mathcal{N}(0, \sigma^2)$.)
   (a) Show that $\lambda_K(\boldsymbol{P}) \geq C\lambda_K(\boldsymbol{Z}^\top\boldsymbol{Z})\lambda_K(\boldsymbol{B})$.
   (b) Using the relationship between $\boldsymbol{U}$ (the eigenvectors of $\boldsymbol{P}$) and $\boldsymbol{Z}$ from the previous homework, given an estimate of the matrix $\boldsymbol{U}_{\mathsf{pure}}$, suggest a procedure to estimate $\boldsymbol{Z}$ from $\boldsymbol{U}$.
   (c) Suppose we are given a black-box algorithm called "vertex hunting" that outputs $\widehat{\boldsymbol{U}_{\mathsf{pure}}}$ such that
   $$\|\widehat{\boldsymbol{U}_{\mathsf{pure}}} - \mathcal{P}^\top\boldsymbol{U}_{\mathsf{pure}}\mathcal{O}\| \leq C\varepsilon.$$
   Here $\mathcal{P}$ is a permutation matrix, and $\mathcal{O}$ is the Frobenius-optimal matrix aligning $\hat{\boldsymbol{U}}$ and $\boldsymbol{U}$. Derive a bound of the form
   $$\|\hat{\boldsymbol{Z}} - \boldsymbol{Z}\mathcal{P}\|_{2,\infty}$$
   in terms of $\varepsilon$, $\lambda_K(\boldsymbol{Z}^\top\boldsymbol{Z})$, $\sigma$, $n$, and $\lambda_K(\boldsymbol{B})$ by relating $\boldsymbol{Z}$ to $\boldsymbol{U}$ as in the last homework and the existing bound on $\|\hat{\boldsymbol{U}}\mathcal{O} - \boldsymbol{U}\|_{2,\infty}$ from class.
   (d) What conditions are needed for consistency as $n \to \infty$?

2. Suppose we observe the matrix
   $$\boldsymbol{M} = \lambda\boldsymbol{u}\boldsymbol{u}^\top + \boldsymbol{N} \in \mathbb{R}^{n\times n}.$$
   where $\boldsymbol{N}$ satisfies
   $$\boldsymbol{N}_{ij} \sim \begin{cases} \mathcal{N}(0, \frac{\sigma^2}{2}) & i < j; \\ \mathcal{N}(0, \sigma^2) & i = j; \\ \boldsymbol{N}_{ji} & j < i. \end{cases}$$
   We assume that $\lambda$ is some large number and $\boldsymbol{u}$ is a unit vector in $n$ dimensions. Prove the following.
   (a) Define the parameter $\theta = \lambda\boldsymbol{u}\boldsymbol{u}^\top$. Show that the MLE for the parameter $\theta = \lambda\boldsymbol{u}\boldsymbol{u}^\top$ over the parameter space of all $n \times n$ matrices satisfies
   $$\hat{\theta} = \boldsymbol{M};$$
   that is, the MLE is the observation itself when no restrictions are placed on the parameter space.
   (b) Show that the mean square error for $\hat{\theta}$ above is $O(n^2\sigma^2)$.
   (c) However, if one instead considers the parameter space $\Theta = \{\text{rank 1 matrices}\}$, then the MLE satisfies
   $$\hat{\theta} = \boldsymbol{M}_1,$$
   where $\boldsymbol{M}_1$ is the best rank one approximation to $\boldsymbol{M}$. (Hint: use the Eckart-Young Theorem)

(d) Show that with high probability it holds that $\|\hat{\theta} - \lambda \boldsymbol{u}\boldsymbol{u}^\top\|_F \leq C\sigma\sqrt{n}$, where $\hat{\theta} = \boldsymbol{M}_1$. (Hint: use the high-probability bound on $\|\boldsymbol{N}\|$ from the last homework. **Note: Make sure to use the correct noise model!**)

(e) Let $\hat{\boldsymbol{x}} = \sqrt{\hat{\lambda}_1}\hat{\boldsymbol{u}}$ denote the leading eigenvector of $\boldsymbol{M}$ scaled by the square root of its corresponding eigenvalue. Let $\boldsymbol{x} = \sqrt{\lambda}\boldsymbol{u}$ be defined similarly. Show that $\hat{\boldsymbol{x}}$ is the MLE for $\boldsymbol{x}$.

(f) Suppose that the leading eigenvalue is known and show that the leading eigenvector $\hat{\boldsymbol{u}}$ is the MLE for the parameter $\boldsymbol{u}$. You may assume that the leading eigenvalue is positive, which happens almost surely for $n$ sufficiently large. (Hint: use the Courant minimax principle.) What if the leading eigenvalue is not assumed known?

(g) Define the loss function

$$f(\boldsymbol{x}) := \frac{1}{4}\|\boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{M}\|_F^2.$$

Compute $\nabla f(\boldsymbol{x})$ and $\nabla^2 f(\boldsymbol{x})$.

(h) Show using the necessary conditions for a local minimum that $\hat{\boldsymbol{x}}$ as in part $(e)$ is a local minimum of this loss function. Is it a global minimum?

(i) Show that since $\hat{\boldsymbol{x}}$ is a global minimum, it must satisfy

$$\|\hat{\boldsymbol{x}} - \boldsymbol{x}\| \leq \frac{C\sigma\sqrt{n}}{\sqrt{\lambda}}$$

provided that $\lambda$ is sufficiently large (as a gauge, it suffices to have $\lambda \geq C_0\sigma\sqrt{n}$ for some large constant $C_0$ but I am okay with stronger conditions. The weakest possible condition is $\lambda > \sigma\sqrt{n}$).